VCF 9.x triage cheat sheet

Work down the steps in order. Each one has a command, what a good result looks like, and what to do when it is not good. Commands are read-only unless marked. Replace `<…>` with your values.

Which layer owns it?

	Management component	Core infrastructure
What	VCF Operations, VCF Automation, identity broker, Ops for networks, log management, services runtime	vCenter, ESX, NSX Managers and Edges, vSAN, SDDC Manager
Tasks	Build > Lifecycle > VCF Management > Tasks	SDDC Manager Tasks, Build > Lifecycle > VCF Instances
Errors	`VCF_LCM_…` with a reference ID, `VCFMS-`, `VCFA-`	Enum codes with a reference token
Logs	Runtime pods in `vcf-fleet-lcm`, `vcf-sddc-lcm`	`/var/log/vmware/vcf/` on SDDC Manager

The steps

1. **Capture the IDs**
   Task details: SDDC Manager Tasks, or VCF Operations Build > Lifecycle > VCF Management > Tasks (9.1)

   ```
   date -u    # logs are in UTC, the UI shows local time
   ```

   GOOD The ticket has the task ID, the reference ID or token, the failed subtask, and the time in UTC.
   IF NOT Do not retry yet. A retry creates a new task and the first failure becomes harder to find.

2. **Read the cause**

   ```
   grep -Ril "<reference-id>" /var/log/vmware/vcf/
   grep -n -B 20 -A 60 "<reference-id>" /var/log/vmware/vcf/<service>/<service>.log | less
   ```

   GOOD The last "Caused by" names something concrete: a host, an account, a certificate, a file system.
   IF NOT Search by the UTC minute instead of the ID. For a management component, read the runtime pod logs (toolkit below).

3. **Check for locks and stale credentials**

   ```
   /opt/vmware/sddc-support/sos --health-check --domain-name <domain>
   ```

   GOOD No active locks, and no "Failed to get details" against an account.
   IF NOT Lock with nothing running: cancel the owner task (API below), fix why it failed, then retry. "Failed to get details": remediate with the password that works on the component today.

4. **Check DNS**

   ```
   ip=$(dig +short <fqdn> | tail -1); dig +short -x "$ip"
   ```

   GOOD The reverse record returns the same FQDN, in lowercase. Run it from the component that failed.
   IF NOT Fix the record or its case, wait for the TTL, then Retry Precheck.

5. **Check time**

   ```
   chronyc tracking 2>/dev/null | grep -E 'System time|Leap' || ntpq -p
   date -u
   ```

   GOOD Every node uses the same source and shows the same UTC time to within a second or two.
   IF NOT Fix NTP first: token errors, 401s and stalled upgrades follow drift. After an NTP change, confirm the setting on the appliance itself.

6. **Check disk**

   ```
   # SDDC Manager
   df -h /data /opt /var/log 2>/dev/null
   # vCenter
   df -h /storage/log /storage/seat /storage/db /storage/core
   ```

   GOOD SDDC Manager: more than 10 GB free in /opt and more than 1 GB for the database (the precheck's critical limits). vCenter: no partition close to full.
   IF NOT Collect a support bundle before you clean anything. Then use the documented clean-up (bundle_cleanup.py for old bundles), not rm.

7. **Check the path**

   ```
   curl -skv --max-time 10 https://<target>/ -o /dev/null 2>&1 | grep -E "Connected|SSL|HTTP/"
   ```

   GOOD Connected, a TLS handshake, an HTTP status. Test from the address that makes the call: in 9.1, often a runtime node.
   IF NOT Timeout: firewall or route. TLS error: inspection proxy or untrusted CA. 403: allowlist or token.

8. **Retry the smallest thing**
   Retry Precheck, or Retry Task: fleet and SDDC lifecycle tasks resume at the failed step. These commands change state.

   ```
   # TOKEN: accessToken from POST https://localhost/v1/tokens (SDDC Manager admin)
   # Retry a failed task
   curl -sk -X PATCH -H "Authorization: Bearer $TOKEN" https://localhost/v1/tasks/<id>
   # Cancel a task that will not complete
   curl -sk -X DELETE -H "Authorization: Bearer $TOKEN" https://localhost/v1/tasks/<id>
   # Cancel a failed password task
   curl -sk -X DELETE -H "Authorization: Bearer $TOKEN" \
     https://localhost/v1/credentials/tasks/<id>
   ```

   GOOD The task resumes and passes the failed step.
   IF NOT Same error again: back to step 2, the cause is not fixed. Password tasks are cancelled and run again, never retried with the generic API.

Symptom, first check, next action

You see	Check	Then
UnknownHostException, installer DNS validation, 502 in the installer	`dig +short <fqdn>; dig +short -x <ip>`	Add or fix the record in lowercase, then Retry Precheck.
"JWT token is expired", IAM_401_INVALID_TOKEN, intermittent 401	`chronyc tracking` on every node	Fix the time source, then retry. Rotating tokens before time is right does nothing.

You see	Check	Then
"Resource is already locked", "Error in acquiring resource locks"	`sos --health-check; GET /v1/tasks/<owner>`	Cancel the owner task, fix why it failed, retry. Delete a lock row only with a snapshot and the lock KB.
Account locked, InvalidLogin, 403 in operationsmanager.log	`sos --health-check (Failed to get details)`	Unlock at the component, find what still holds the old password, then Remediate. Rotate only after it is healthy.
PKIX path building failed, certificate_unknown(46)	`openssl s_client -connect <host>:443 </dev/null | openssl x509 -noout -issuer -enddate`	Expired: replace. Unknown issuer: TLS inspection on the path, exempt it, or import the CA.
Signed certificate rejected	`openssl x509 -in cert.crt -pubkey -noout | sha256sum; openssl req -in csr.csr -pubkey -noout | sha256sum`	Hashes differ: the certificate was signed from another CSR. Sign the current one.
"no healthy upstream" on vCenter	`df -h; service-control --status --all; the VECS expiry loop`	Free the full partition or replace the expired certificate, then start services. Rebooting alone ends in the same place.
VCFMS- precheck codes, 9.1 runtime tasks stuck	`kubectl get pods -A | grep -vE 'Running|Completed'`	Fix pod health first. Restart a lifecycle service with lcm_service_support.sh, not kubectl delete.
Depot 403, "Bundle not found", download stalls	`curl -I https://eapi.broadcom.com from a runtime node`	Allow every runtime node, not only SDDC Manager. Check the token or activation code, and TLS inspection.
NSX host preparation or TEP failures	`vmkping ++netstack=vxlan -d -s 1572 -I vmk10 <edge-tep>`	Fails with -d: MTU on the path is under 1600. Fix the network, then Resolve in NSX.
VCF Operations slow, node offline, cluster not ready	`curl -sk https://<node>/casa/cluster/status; df -h /storage/db /storage/log`	One node must be MASTER. Add disk to every node, identically sized. Never revert to a snapshot.

Log map

SDDC Manager	`/var/log/vmware/vcf/` : `lcm/lcm-debug.log` (prechecks, bundles, upgrades), `domainmanager/` (domains, clusters, hosts), `operationsmanager/` (passwords, certificates, backups, commissioning), `commonsvcs/` (tokens, SSH keys, API)
VCF Operations	`/storage/log/vcops/log/` : `collector.log` , `web-vcf.log` , `adapters/<Kind>/` , `casa/casa.log`
Runtime (9.1)	Fleet: `vcf-fleet-lcm` . SDDC: `vcf-sddc-lcm` . Depot: `vcf-fleet-depot` . Platform: `vmsp-platform`
vCenter	`/var/log/vmware/` : `vsphere-ui/logs/` , `cis-license/license.log` , `vmware-updatemgr/`
NSX	`/var/log/cm-inventory/cm-inventory.log` , `/var/log/syslog`

Collect before you escalate

```
cd /opt/vmware/sddc-support
# One workflow failed
./sos --sddc-manager-logs --zip
# The failure spans hosts, vCenter or NSX
./sos --collect-all-logs --zip --domain-name
<domain>
# 9.1 management services as well
./sos --sddc-manager-logs --vcf-mgmt-svcs-logs --zip
```

- **VCF Operations:** Control Panel > Support Bundles.
- **Runtime down, fleet lifecycle unreachable:** the standalone bundle script (KB 444466).
- **UI problems:** a browser HAR file.
- **Always:** the IDs from step 1, the UTC time, and every action taken since.

Rules

One operation at a time on a scope. Health before retry, retry before rebuild. Bundle before clean-up. Snapshot before any database change, and make database changes only when a KB for your exact build says to.

Services runtime toolkit (9.1, read-only)

```
# Control-plane node IP: Build > Lifecycle > Components > VCF Services Runtime > Nodes
ssh vmware-system-user@<control-plane-node>
sudo su -
export KUBECONFIG=/etc/kubernetes/admin.conf
kubectl get nodes -o wide
kubectl get pods -A | grep -vE 'Running|Completed'
kubectl get events -n vcf-fleet-lcm --sort-by=.lastTimestamp | tail -30
P=$(kubectl get pods -n vcf-fleet-lcm -l app=fleetbuild -o jsonpath='{.items[0].metadata.name}')
kubectl logs $P -n vcf-fleet-lcm --tail=200
kubectl -n vcf-fleet-lcm exec vcf-fleet-lcm-db-0 -- patronictl list  # one leader, no lag

# Changes. Stuck fleet or SDDC lifecycle task (KB 435569): report, then restart one service
lcm_service_support.sh -t
lcm_service_support.sh -r fleetbuild    # or fleetupgrade, sddcbuild, sddcupgrade
```

Find `lcm_service_support.sh` and run it as KB 435569 describes for your build. Never delete pods or `vcf-svc-*` VMs to restart them. More in the services runtime for VI admins.

Not official vendor guidance. Confirm against current Broadcom documentation for your build before changing production.